

2024 Calculus II Supplement Homework (4)

1. Exercise 42, (1),(3),(5),(7) and Exercise 43,(1) in page 28 from Calculus Supplement Homework.

2. Exercise 53 in page 29 from Calculus Supplement Homework.

3.(1). Express the function $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ as the Maclaurin series.

(2). Express the function $f(x) = x^2 - 5x + 6$ as the Taylor series at $x = 4$.

(3). Express the function $f(x) = \frac{1}{x^2+3x+2}$ as the Maclaurin series.

(4). Express the function $f(x) = \frac{1}{x^2-5x+6}$ as the Taylor series at $x = 4$.

4. Find the following limits.

(a) $\lim_{x \rightarrow 0} \left(\frac{1}{\ln(1+x)} - \frac{1}{e^x \sin x} \right).$ (b) $\lim_{n \rightarrow \infty} \left[(n^2 - n)e^{\frac{1}{n}} - \sqrt{n^4 + 1} \right].$

5 Let $f(x) = \tan x$. Find $f^{(5)}(0)$.